

Does a CNN Rely More on Global Shape or Local Texture?

A cue-conflict study with a fine-tuned ResNet-50 on a 4-class animal dataset

Research Question

When global shape and local texture provide conflicting class cues, which cue does our trained ResNet-50 rely on more during classification?

Prediction (written before the main cue-conflict experiment)

We predicted that the trained ResNet-50 would rely more strongly on local texture than global shape under the cue-conflict setup. This prediction was fixed before the main cue-conflict experiment and was not revised after seeing the experimental results.

Dataset

Animal Computer Vision Clean Dataset and Code (Emirhan Bulut, Kaggle). 4,000 images, perfectly balanced across 4 classes (Buffalo, Elephant, Rhino, Zebra; 1,000 each), verified by direct file count. Link: kaggle.com/datasets/emirhanai/animal-computer-vision-clean-dataset-code-cnnai. License: **CC BY 4.0** (Creative Commons Attribution 4.0 International), verified directly on the Kaggle dataset page. Attribution: dataset created by Emirhan Bulut, used here under the terms of CC BY 4.0.

Method Overview

- **Split:** 70/15/15 per class, fixed seed (700/150/150 images per class for train/validation/test).
- **Model:** ImageNet-pretrained ResNet-50, fine-tuned with only layer4 and the final classifier head unfrozen; the rest of the network stays at its pretrained weights.
- **Checkpoint selection:** best checkpoint chosen using the validation set only. The test set is never touched until after the checkpoint is frozen.
- **Cue-conflict stimuli:** Gatys-style neural style transfer (VGG19 features, Gram-matrix style loss, LBFGS optimizer, 150 steps) is used to combine the *shape* of one class's photo (content image) with the *texture* of a different class's photo (style image), producing an image whose two cues point to two different classes.
- **Trial design:** all 12 ordered class pairs (shape $\neq$ texture). For each pair, 40 content images (from the held-out *test* split) are each combined with 3 style images (from the *train* split), giving $40 \times 3 = 120$ trials per direction and $12 \times 120 = 1,440$ trials total. Both content and style pools are drawn with a fixed random seed, not hand-picked for outcome.
- **Frozen-model protocol:** the classifier is never retrained or fine-tuned again after the baseline checkpoint is frozen — it is used purely for inference throughout style-transfer generation and cue-conflict evaluation.
- **Classification of each stylized image** as a *shape choice* (prediction matches the content/shape class), a *texture choice* (prediction matches the style/texture class), or *other* (neither).

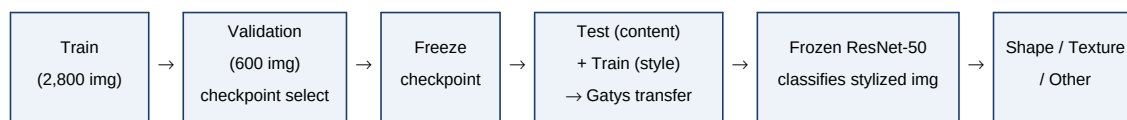


Figure 1. Experimental pipeline, left to right.

Non-independence note: the 1,440 trials are not 1,440 independent samples — each of the 40 content images per direction contributes 3 correlated trials (one per style reference). We therefore report results at both the *trial level* and the *content-image level* (3 style trials per content image averaged first, then averaged across the 40 content images); see Results.

Baseline: Can the Model Classify Normal Images First?

Before any cue-conflict interpretation is meaningful, the fine-tuned model must first classify ordinary, unmodified photographs well. We evaluated the frozen checkpoint once on the untouched 600-image test set (150/class).

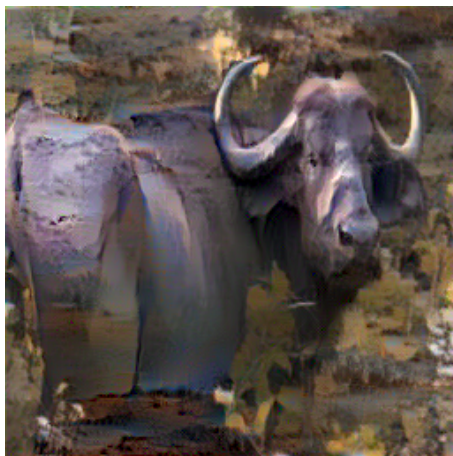
Metric	Value
Test accuracy	98.83%
Test macro-F1	98.83%
Buffalo (precision / recall / F1)	0.987 / 0.987 / 0.987
Elephant (precision / recall / F1)	0.993 / 1.000 / 0.997
Rhino (precision / recall / F1)	0.974 / 0.987 / 0.980
Zebra (precision / recall / F1)	1.000 / 0.980 / 0.990

Confusion matrix (rows = true, columns = predicted; order Buffalo, Elephant, Rhino, Zebra): [148,0,2,0] / [0,150,0,0] / [1,1,148,0] / [1,0,2,147].

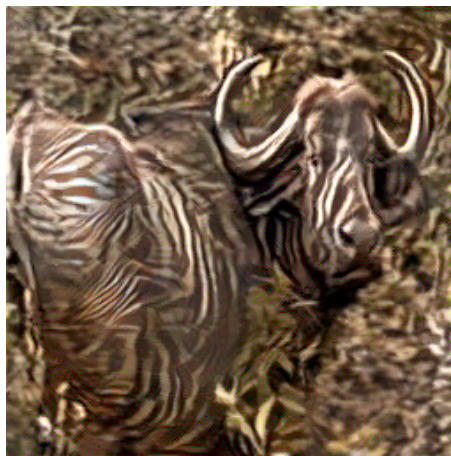
Assessment: every per-class F1 ≥ 0.98 with a near-diagonal confusion matrix — the model classifies all four classes well on natural images. This clears the baseline acceptance criterion, so the cue-conflict results below are interpretable and not an artifact of a poorly trained model.

Cue-Conflict Stimuli

The figure below shows two example stylized images: a typical direction (Buffalo shape + Elephant texture) and one of the three Zebra-as-texture directions discussed in Results, where the stripe texture is visually much stronger.



(a) shape=Buffalo, texture=Elephant — predicted Buffalo (shape choice, typical direction)



(b) shape=Buffalo, texture=Zebra — one of the low-shape-choice directions (68.3% shape here, vs. 87.4% overall)

Figure 2. Two representative cue-conflict stimuli generated with Gatys-style style transfer. The full set contains one example from each of the 12 directions (in examples/).

Important caveat on “texture”: neural style transfer does not isolate texture with perfect purity — it can carry along color and some local structure from the style image. We therefore treat the style-transfer condition as constructing a texture-defined cue (a style-defined visual cue built to conflict with shape), not as a pure, physically isolated measurement of texture reliance. Results are interpreted accordingly throughout this report.

Main Result: Cue-Conflict Experiment

Across all 1,440 trials, the model overwhelmingly selected the **shape**-defined class over the texture-defined class:

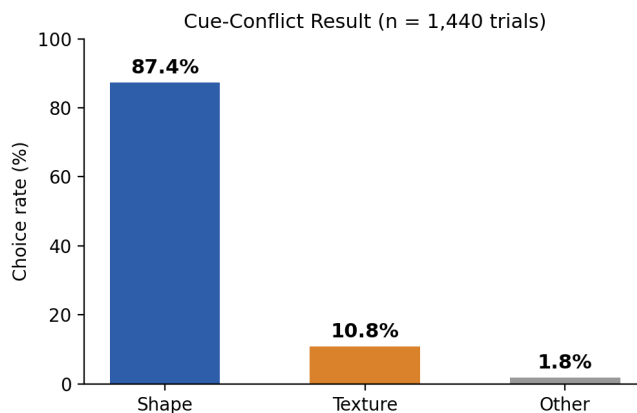


Figure 3. Trial-level Shape / Texture / Other choice rates, $n = 1,440$. Content-image-level aggregation (3 style refs per content image averaged first, then across 40 content images per direction) gives essentially identical rates (87.4% / 10.8% / 1.8%), indicating the pattern is consistent within each content image and not driven by any single style reference.

Result vs. prediction: the hypothesis was NOT supported. We predicted texture would dominate; instead, *under our cue-conflict experimental setup, the trained ResNet-50 selected the shape-defined class in 87.4% of trials*, versus 10.8% for the texture-defined class and 1.8% for neither (average confidence 0.922). We did not revise the prediction after seeing this result.

Not Uniform Across Class Pairs: the Zebra Exception

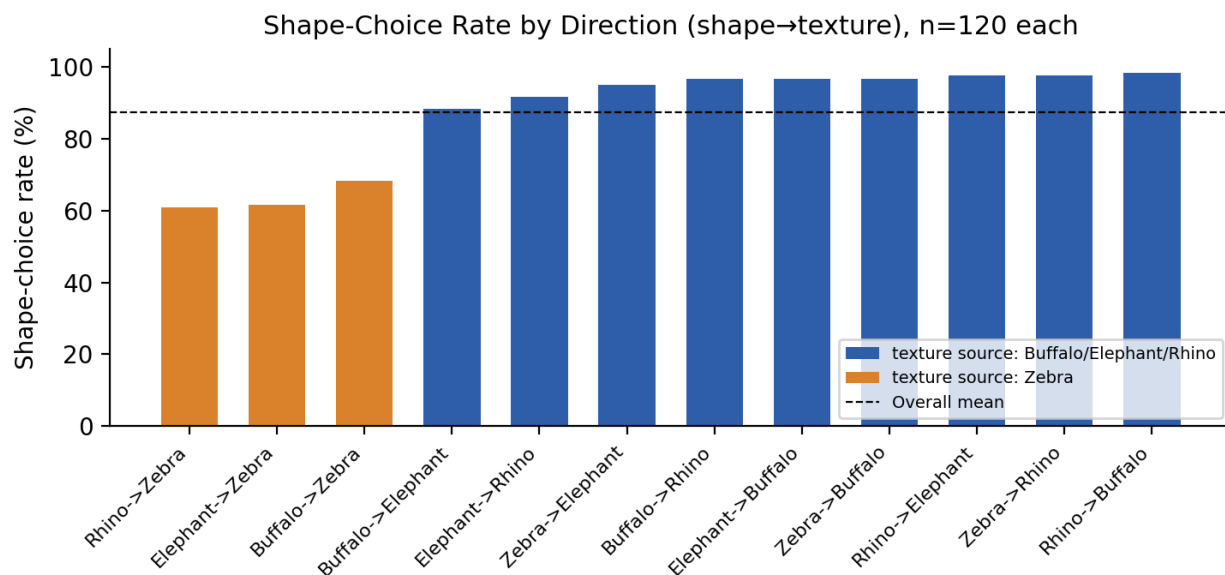
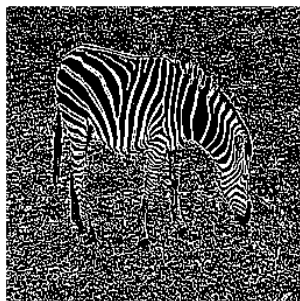


Figure 4. Shape-choice rate for each of the 12 directions ($n=120$ each), sorted ascending. Dashed line = overall mean (87.4%).

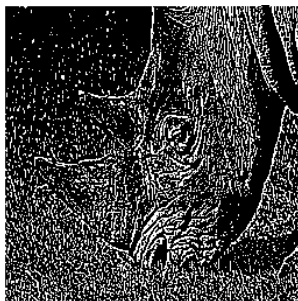
Although shape was dominant overall, the effect was not uniform across class pairs. In particular, the three directions using **Zebra as the texture source** (Buffalo→Zebra 68.3%, Elephant→Zebra 61.7%, Rhino→Zebra 60.8%) show substantially lower shape-choice than every other direction (88.3%–98.3%). By contrast, when Zebra supplies the *shape* instead (Zebra→Buffalo/Elephant/Rhino), shape-choice is back to normal-to-high (95.0%–96.7%). We present this as an observation rather than a firm causal claim: zebra stripes appear to be an unusually strong texture-defined cue specifically when acting as the style/texture source, but we did not run a follow-up experiment to isolate exactly why.

Supporting Control: Edge-Based Shape Representation

As a supporting check, we took the same 40 test-content images per class already used in the main experiment (160 images total, identical seeding) and converted each into an edge-based representation using PIL's `FIND_EDGES` filter followed by thresholding, substantially reducing color information while retaining intensity boundaries and local edges, then classified them with the same frozen checkpoint — no style transfer or retraining involved. Note that `FIND_EDGES` detects intensity transitions generally; it does not separate an object's outer contour from edges produced by internal texture (visible, for example, in the Zebra example below, where stripe edges remain prominent) — so this condition is an edge-based structural representation, not a clean isolation of shape with texture fully removed.



Zebra edge map (100% correct) — note stripe edges remain visible



Rhino edge map (37.5% correct)

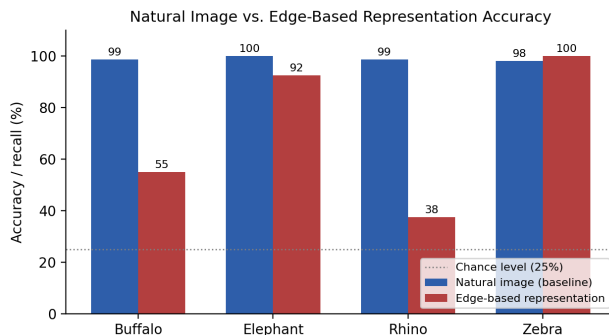


Figure 5. Baseline vs. edge-representation accuracy, overall and per class.

Overall: 71.3% accuracy (114/160), average confidence 0.581, versus 98.8% on natural images and 25% chance for 4 classes.
Per-class: Zebra 100%, Elephant 92.5%, Buffalo 55.0%, Rhino 37.5%.

Caveat (must be read together with the number above): this edge-based representation is out-of-distribution relative to the natural-image training data, so this result is treated as supporting evidence only, not as a second independent confirmation of equal strength to the cue-conflict experiment — the 27.6-point drop from baseline may reflect the distribution shift and the loss/transformation of multiple visual cues, rather than the removal of texture alone.

With that caveat in place, the edge-based representation still supports classification far above chance, which suggests that boundary and structural information carries substantial class-relevant information for this model. The per-class pattern is also informative: Zebra's edge map is classified perfectly even though its stripe edges remain visible in the representation, while Rhino and Buffalo edge maps are considerably harder to classify — how well this edge-based representation works as a cue is not uniform across classes.

Interpretation

We predicted the trained ResNet-50 would favor the texture-defined cue; instead it showed a **strong shape preference** under the cue-conflict setup (87.4% shape vs. 10.8% texture), corroborated by the edge-based representation still reaching 71.3% accuracy, far above chance. One plausible explanation is that this visually simple four-class task, combined with fine-tuning only the later layers (Layer 4 and the classifier head) of an ImageNet-pretrained model, may favor global outline cues such as body shape, horns, or ears. The result is not uniform, however — zebra stripes specifically pull shape-choice down to 61–68% when used as the texture source, suggesting that the Zebra style-defined cue can substantially influence the model when it acts as the texture source.

Limitations

- **Scope:** these findings describe this specific trained ResNet-50, on this dataset, with this training procedure, this style-transfer method, and this cue-conflict setup. They should not be generalized to “CNNs” as a class of models or to other architectures, datasets, or training regimes.
- **“Texture” is not perfectly isolated.** Gatys-style style transfer can carry along color and some local structure from the style image, so the texture condition is more accurately described as a texture-defined (style-defined) cue rather than a pure measurement of texture reliance.
- **1,440 trials are not 1,440 independent samples** — 120 trials per direction come from only 40 unique content images $\times$ 3 style images. We mitigated this by reporting both trial-level and content-image-level results (which agreed closely), but true independent-sample count is smaller than 1,440.
- **The edge-based control is out-of-distribution** relative to natural-image training data, and **FIND_EDGES does not cleanly isolate shape** — it detects intensity transitions generally, so edges from internal texture (e.g. zebra stripes) remain visible alongside the outer contour. This condition is therefore supporting evidence for the role of boundary/structural information, not a clean, confound-free second measurement of shape alone.
- **Only 4 classes, one dataset, one architecture (ResNet-50), one style-transfer method (Gatys/VGG19).** Whether the shape-preference finding holds for other class sets, architectures, or texture-generation methods is untested here.
- **Per-class asymmetry** (e.g. Rhino edge-representation accuracy 37.5% vs. Zebra 100%) shows reliance on this cue is not uniform across classes — the overall 87.4%/71.3% figures average over meaningfully different per-class behavior.

Reproducibility

All randomization (data split, content/style sampling, model initialization) uses a fixed seed (42), recorded in code. Pilot development images (used only to select the style-transfer method and step count) are drawn from an explicitly excluded index range and never appear in the 1,440 cue-conflict trials or the 160 edge-representation control images. Code, per-trial JSONL logs (`cue_conflict_trials.jsonl`, `silhouette_trials.jsonl`), summary JSON files, the trained checkpoint, and this report are included in the submitted repository so all reported numbers can be regenerated.

AI Usage Disclosure

AI assistance (Claude) was used in this project in the following ways:

- **Experimental design:** helped design and iteratively refine the cue-conflict protocol (dataset verification, train/validation/test data-role separation, seeded sampling, pilot-vs-experimental-data separation, trial/content-level non-independence handling, edge-representation control design).
- **Interpretation planning:** helped plan how to analyze and present results honestly, including scoping conclusions to this specific model/dataset/setup and flagging the texture-defined-cue and edge-representation-OOD caveats used throughout this report.
- **Writing assistance:** helped draft and structure this report from the experiment's own numeric outputs.
- **Where AI needed correction:** during pilot development, the AI initially misdiagnosed a noisy AdaIN-proxy output as a failed weight download, when the actual cause (identified by the student, who had written the pilot script) was an unregularized Adam pixel-space optimization. The AI's early framing was corrected before proceeding, and the project switched to a full Gatys optimization method as a result. All experimental code that produced the reported numbers (baseline training, main cue-conflict experiment, results analysis) was written and executed by the student; the silhouette control script was AI-written but reviewed and run by the student.